Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/25/2005
PI Name:	Hlatky, Lynn Ph.D.		
Project Title:	Computational Modeling of Chromosome Aberrations Produced by HZE Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02135-2907	**Congressional District:**	8
Comments:	PI affiliation name change to Steward Research and Specialty Projects Corporation (2012). PI formerly at Caritas St. Elizabeth's Medical Center, Boston, MA (2005-2011); Dana-Farber Cancer Institute, Boston, MA (prior to 2005). Subsequent name change from Steward Research and Specialty Projects Corporation to GeneSys Research Institute in 2013.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	06/01/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hahnfeldt, Philip (Caritas St. Elizabeth's Medical Center, Tufts Univ School of Medicine) Sachs, Rainer (University of California)		
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Task Description:

Chromosome aberrations produced by high LET radiations are directly and indirectly relevant to space exploration and risk estimation in a number of different ways. They are important as probes for DNA repair and recombination mechanisms, as biomarkers of dose and radiation quality, as factors in genomic instability and carcinogenesis, and as surrogates for cancer in risk estimation. Quantitative modeling is needed to exploit and augment the rapidly accumulating empirical data on aberration types and frequencies. This project will apply mathematical and computer algorithms to data on aberrations produced by iron ions (56Fe). The analysis will be compared to analyses of damage produced by photons, electrons, or alpha particles, with the goal of assessing signature responses to 56Fe, an important radiation encountered in space. To perform such studies, the team has tied damage endpoints to risk through our customized software and algorithms. These endpoints include: 1. aberration complexity, quantified by cycle structures; 2. apparently and truly incomplete aberrations, related to segment size distributions; 3. aberration spectra as related to chromosome geometry, to DNA double strand break (DSB) clustering depending on radiation track structure, and especially to DSB misrejoining depending on proximity; 4. mFISH color junctions for each specific chromosome pair, related to chromosome-chromosome spatial correlations in an interphase nucleus.
Substantial preliminary studies have recently been carried out by our team for each aim. As remarked, we will employ the team's specialized and comprehensive software, including the highly-developed chromosome aberration simulator (CAS) software, for analyzing the aberration spectra encountered. The data to be analyzed will be for human cells in vitro, scored by mFISH or other similar techniques. A number of these data sets are being made available by many of our colleagues worldwide. This will greatly expand the range of suitable data available for analysis. We have also recently devised methods for systematically comparing mFISH data with data obtained by older methods such as FISH or solid staining and with newer methods such as color banding.

For Specific Aim 1, we will replace the current approach of using "obligate" (deterministic) cycles with a more accurate computer-based approach involving cycle probabilities. For Aim 2, new methods are being developed that allow systematic comparison of apparently incomplete aberrations with corresponding complete aberrations and corresponding truly incomplete ones, even in the case of very complex aberrations. Under Aim 2 we will also consider size distributions of fragments that have one telomere and one unrejoined free end, for comparison to recent data using pan-telomeric probes; and analyze mixtures of different radiations. Aim 3 will emphasize more realistic computer algorithms for break misrejoining, bringing this aspect of aberration modeling more nearly in line with the very sophisticated models now being used by various groups for assessing the influence of track structure and chromatin geometry on DSB locations. For Aim 4 we plan to apply a method we previously used for low LET data to data for high LET, including iron ions. The method involves testing frequencies of color junction pairs for deviations from randomness.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

For FY 2005: We carried out specific aim number 4: statistically analyzing mFISH color junction frequencies for radiation-produced chromosome aberrations. We used the pairwise frequencies to indicate chromosome-chromosome spatial associations related to the excess production of interchanges and also used the color junction frequencies to indicate the radiosensitivity of each chromosome. Data and results on high LET radiation (i.e. densely ionizing radiation such as alpha particles, or HZE particles) were added to the analyses for low LET (i.e. for low linear energy transfer, corresponding to sparsely ionizing radiation such as hard x-rays or gamma rays). The high LET data is particularly pertinent to the kind of radiation encountered in an extended space voyage. We found that proximity effects on interchanges due to geometric associations between chromosomes in an interphase nucleus are more pronounced at high LET.
A previously suggested chromosome cluster in lymphocytes consisting of 5 gene-rich chromosomes was confirmed and a cluster containing the nucleolus chromosomes was confirmed. Chromosome associations and preferential radiogenic interchanges due to proximity effects are important in radiation carcinogenesis, especially radiation leukemogenesis. The results on specific aim number 4 were reported in a publication in the journal "Bioinformatics" and are available using an interactive Java program.

For the part of aim number 1 dealing with analyzing DNA damage/repair/misrepair pathways we used mFISH results on radiogenic homologue-homologue dicentrics, including results on the HZE radiation emphasized by NASA, to show that there is a dependence on homology even for mammalian cells in the G0/G1 phase of the cell cycle. There is a highly significant excess of one color (i.e. homologue-homologue) dicentrics compared with the number of such dicentrics expected from two-color (i.e. heterologous) dicentrics and computer models or analytic estimates. This homology dependence could be due to geometric associations between homologues in the interphase nucleus or to homology-dependent misrepair processes. The dependence of the observed excess on LET is not very strong. The results are reported in a paper accepted for publication in a peer-reviewed journal, the International Journal of Radiation Biology.

We made very substantive progress on that part of aim 2 which deals with systematically comparing apparently incomplete, truly incomplete, and complete aberrations. In brief, the optimal procedure is identify incompleteness type by looking for mismatches between centromeres and apparent telomeres involved in a particular aberration since in a complete aberration there are two telomeres of each color for each centromere of that color. Once incompleteness type has been characterized, color by color, adding "virtual fragments" of appropriate colors gives a fully methodical way to see what the possible completions are. The results are being written up. An applet developed during preliminary studies, already available on the internet at the url http://radiobiology.math.berkeley.edu/ , is being updated with new algorithms that give a more realistic description of the probabilities for different complete aberrations that can correspond to a given apparently incomplete mFISH aberration; previously, apparent incompleteness has been a major confounding factor in analyzing HZE data on aberrations. We continue to make progress on aim 3, computer simulations of aberration formation. We have developed two geometric approaches to high LET radiation simulation. One is based on random consecutive location of chromosomes and uses our randomly located cluster formalism; the other is based on random walk geometry for all 46 chromosomes in an interphase nucleus and uses track structure simulations. The former has recently been implemented by Java software involving about 30 classes and interfaces. This code is documented in detail at the url http://math.berkeley.edu/ . Testing of the program and using it for analysis of high LET (specifically alpha particle) radiation data are underway.

Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Plan Y, Hlatky L, Hahnfeldt P, Sachs R, Loucas B, Cornforth M. "Full-color painting reveals an excess of radiation-induced dicentrics involving homologous chromosomes." Int J Radiat Biol. 2005 Aug;81(8):613-20. PMID: 16298942 , Aug-2005
Articles in Peer-reviewed Journals	Vives S, Loucas B, Vazquez M, Brenner DJ, Sachs RK, Hlatky L, Cornforth M, Arsuaga J. " SCHIP: statistics for chromosome interphase positioning based on interchange data." Bioinformatics. 2005 Jul 15;21(14):3181-2. Epub 2005 May 10. PMID: 15886279 , Jul-2005